Quantitative Aptitude

A Comprehensive Guide for Competitive Examinations

Chapter 1 — Number System

The number system forms the foundation of all quantitative reasoning. In this chapter we cover natural numbers, whole numbers, integers, rational numbers, irrational numbers, and real numbers. A thorough understanding of these concepts is essential for solving problems in arithmetic, algebra, and number theory that frequently appear in competitive examinations such as the GRE, GMAT, CAT, and various banking recruitment tests.

1.1 Classification of Numbers

Natural numbers are the counting numbers 1, 2, 3, 4, and so on. They are the most basic class of numbers and are denoted by the symbol N. Whole numbers extend natural numbers by including zero, giving the set {0, 1, 2, 3, ...}. Integers include all whole numbers together with their negative counterparts, written as Z = {..., -3, -2, -1, 0, 1, 2, 3, ...}. Rational numbers are those that can be expressed as a fraction p/q where p and q are integers and q is not zero. Irrational numbers, such as $\sqrt{2}$ and π, cannot be expressed as such fractions and have non-terminating, non-recurring decimal expansions.

Example 1:

> Classify the following numbers: -7, 0, 3/4, $\sqrt{5}$, 2.5, π.

Solution: -7 is an integer (and also a rational number since it equals -7/1). 0 is a whole number. 3/4 is a rational number. $\sqrt{5}$ is irrational because it cannot be written as a fraction of integers. 2.5 is rational because it equals 5/2. π is irrational. Notice that every integer is also a rational number, but not every rational number is an integer.

Chapter 2 — Percentages

Percentage is one of the most widely tested topics in quantitative aptitude. The word "percent" comes from the Latin per centum, meaning "out of one hundred". A percentage represents a fraction with denominator 100. The symbol % is used to denote percentages. Understanding percentages is crucial because they appear in problems involving profit and loss, simple and compound interest, data interpretation, and statistical analysis.

2.1 Basic Formula

If a value x is to be expressed as a percentage of a value y, then: Percentage = (x / y) × 100%

For example, if a student scores 45 marks out of a maximum of 60, the percentage is (45 / 60) × 100% = 75%. This means the student has achieved 75 out of every 100 possible marks. Percentages allow us to compare performance across different scales — a score of 90/120 is the same as 75%, just like 45/60.

2.2 Percentage Increase and Decrease

When a quantity changes from an original value to a new value, the percentage change is given by: Percentage Change = ((New Value - Original Value) / Original Value) × 100%. If the new value is greater than the original, the change is an increase; if it is smaller, the change is a decrease. A common mistake is to confuse the base — the denominator is always the original value, not the new value.

Example 2:

> The price of a commodity increases from $80 to $100. What is the percentage increase?

Solution: Original value = $80, New value = $100. Percentage increase = ((100 - 80) / 80) × 100% = (20 / 80) × 100% = 25%. Therefore, the price increased by 25%. Notice that going backwards (from $100 to $80) would give a 20% decrease, not 25% — the base matters.

Chapter 3 — Time, Speed, and Distance

The relationship between time, speed, and distance is fundamental in physics and arithmetic. The basic formula connecting these three quantities is: Distance = Speed × Time. This means that if you know any two of these values, you can calculate the third. The formula can be rearranged as Speed = Distance / Time or Time = Distance / Speed, depending on what you need to find.

3.1 Units and Conversions

The most common units are kilometres per hour (km/h) for speed, hours or seconds for time, and kilometres or metres for distance. To convert from km/h to m/s, multiply by 5/18. To convert from m/s to km/h, multiply by 18/5. For example, 72 km/h equals 72 × 5/18 = 20 m/s. Conversely, 10 m/s equals 10 × 18/5 = 36 km/h. These conversions are essential when the units in a problem are mixed.

3.2 Relative Speed

When two objects move in the same direction, their relative speed is the difference of their individual speeds. When they move in opposite directions, their relative speed is the sum of their individual speeds. This concept is critical for solving problems involving trains passing each other, boats moving upstream or downstream, and people walking towards or away from each other.

Example 3:

> Two trains, 120 metres and 80 metres long, are running on parallel tracks in opposite directions at 60 km/h and 40 km/h respectively. How long will they take to cross each other completely?

Solution: Total distance to be covered = 120 + 80 = 200 metres. Relative speed (opposite directions) = 60 + 40 = 100 km/h. Convert to m/s: 100 × 5/18 = 250/9 m/s. Time = Distance / Speed = 200 / (250/9) = 200 × 9 / 250 = 7.2 seconds. Therefore, the trains will take 7.2 seconds to completely cross each other.

Chapter 4 — Profit and Loss

Profit and loss problems revolve around the cost price (CP) and selling price (SP) of an item. When the selling price is greater than the cost price, there is a profit. When the selling price is less than the cost price, there is a loss. The fundamental formulas are: Profit = SP - CP, Loss = CP - SP, Profit Percentage = (Profit / CP) × 100%, and Loss Percentage = (Loss / CP) × 100%. Note that percentages are always calculated on the cost price, not the selling price.

4.1 Marked Price and Discount

In retail scenarios, sellers often mark goods at a price higher than the cost price and then offer a discount to attract customers. The marked price (MP) is the price printed on the item. The selling price is then calculated as SP = MP × (1 - Discount/100). The seller's profit depends on the relationship between the cost price and the final selling price after discount.

Example 4:

> A shopkeeper marks an item at $200 and offers a 15% discount. If his profit is 20%, what was his cost price?

Solution: Selling price after discount = $200 × (1 - 0.15) = $200 × 0.85 = $170. Since profit is 20%, the cost price CP satisfies: SP = CP × 1.20, so CP = SP / 1.20 = $170 / 1.20 = $141.67 (approximately). Therefore, the shopkeeper's cost price was about $141.67.

Chapter 5 — Data Interpretation Reference

The following table summarises the most frequently tested aptitude topics and their approximate weight in major competitive examinations. Use this as a quick reference when planning your preparation strategy.

Topic	Approx. Weight (%)	Difficulty	Average Time (min)
Number System	15	Easy	2
Percentages	12	Easy	2
Profit and Loss	10	Medium	3
Time, Speed, Distance	10	Medium	3
Time and Work	10	Medium	3
Ratio and Proportion	8	Easy	2
Average	8	Easy	2
Simple Interest	7	Easy	2
Compound Interest	7	Medium	3
Probability	5	Hard	4
Permutations	5	Hard	4
Geometry	**3**	**Hard**	**5**

This book covers each of the above topics in detail with worked examples, practice problems, and shortcut techniques. The key to mastering quantitative aptitude is consistent practice — aim to solve at least 20 problems per topic per day, and review your mistakes carefully to identify patterns in your errors.

Contact and Resources

For additional practice material, visit our website at https://aptitudeprep.example.com or email us at support@aptitudeprep.example.com. You can also call our helpline at +91-98765-43210 between 9 AM and 9 PM IST, Monday through Saturday. Follow us on social media @aptitudeprep for daily practice questions and tips.

First published in 2024 by Aptitude Prep Publishers. ISBN 978-93-84567-12-3.